



**Department of Mechanical Engineering**  
**Internal Assessment Question paper**

**Programme:** Common to all branches (UG)

**Term:** 04<sup>th</sup> OCT 2023 – 20<sup>th</sup> Jan 2024

**Course:** Design thinking

**Course Code:** AECC17

**CIE:** Test 1    **Semester:** I    **Sec:** M,N,O,P,Q,R,S,T,U,V & W

**Max Marks:** 30

**Time:** 1Hr    **Portions for Test:** Lesson Plan – L1 to L09

**Instructions to Candidates:**

- Answer any 2 out of 3 questions
- Meaningful Sketches to be drawn with pencil and scale only

| Sl. No. | Question's                                                                                                                                            | Marks | Blooms level | Course Outcome |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------------|----------------|
| 1       | a) What is Design Thinking? How does it work?                                                                                                         | 5     | L2           | CO1            |
|         | b) Justify why Critical thinking, Creative Thinking, Communication and Collaboration are essential skillsets for students with suitable example       | 10    | L5           | CO1            |
| 2       | a) With a suitable example briefly explain each step in the Design Thinking process                                                                   | 10    | L2           | CO1            |
|         | b) Briefly explain <ul style="list-style-type: none"><li>i. Immersion</li><li>ii. Research</li><li>iii. Observations</li><li>iv. Interviews</li></ul> | 5     | L2           | CO2            |
| 3       | a) Write 10 interview questions to ask teachers pertaining to problems faced while using chalk and blackboard                                         | 10    | L5           | CO2            |
|         | b) Differentiate between NEEDS (FINDINGS) and INSIGHTS with a suitable example                                                                        | 5     | L3           | CO2            |

**MOBILE PHONES ARE BANNED IN THE EXAM HALL**





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| 1       | a) What is Design Thinking? How does it work?                                                                                                         | 5     | L2           | CO1            |
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| 2       | a) With a suitable example briefly explain each step in the Design Thinking process                                                                   | 10    | L2           | CO1            |
|         | b) Briefly explain <ul style="list-style-type: none"><li>i. Immersion</li><li>ii. Research</li><li>iii. Observations</li><li>iv. Interviews</li></ul> | 5     | L2           | CO2            |
| 3       | a) Write 10 interview questions to ask teachers pertaining to problems faced while using chalk and blackboard                                         | 10    | L5           | CO2            |
|         | b) Differentiate between NEEDS (FINDINGS) and INSIGHTS with a suitable example                                                                        | 5     | L3           | CO2            |

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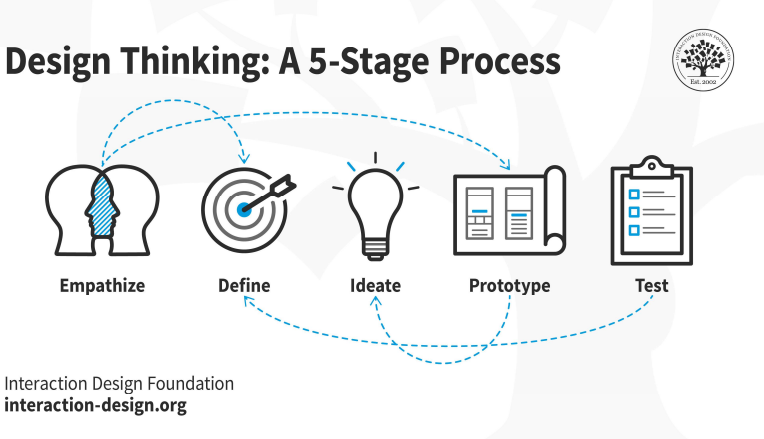


## Department of Mechanical Engineering

### Scheme of Evaluation

#### Instructions to Candidates:

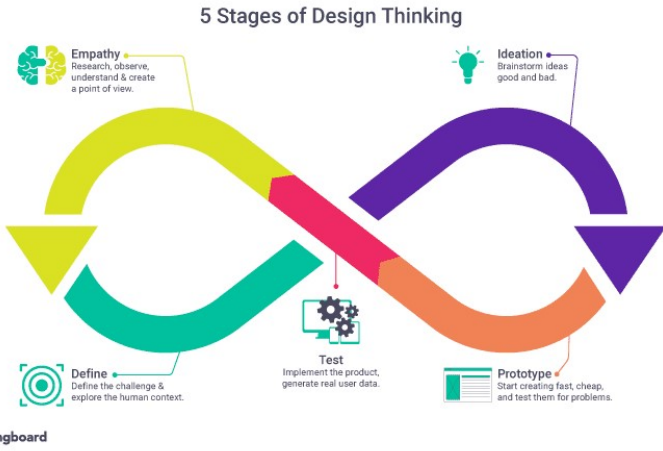
- Answer any 2 out of 3 questions
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| Sl. No. | Question's                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1       | <p>Design thinking is a non-linear, iterative process that teams use to understand users, challenge assumptions, redefine problems and create innovative solutions to prototype and test. Involving five phases—Empathize, Define, Ideate, Prototype and Test—it is most useful to tackle problems that are ill-defined or unknown – <b>3 marks</b></p> <p>Mention the process below – <b>2 marks</b></p> <p>a) <b>Design Thinking: A 5-Stage Process</b></p>  <p>Interaction Design Foundation<br/>interaction-design.org</p> | 5     |



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|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
|  | <p><b>Critical Thinking – 2 marks</b></p> <p>Critical thinking is focused, careful analysis of something to better understand it. When people speak of "left brain" activity, they are usually referring to critical thinking. Here are some of the main critical-thinking abilities:</p> <ul style="list-style-type: none"><li>* Analyzing is breaking something down into its parts, examining each part, and noting how the parts fit together.</li><li>* Arguing is using a series of statements connected logically together, backed by evidence, to reach a conclusion.</li><li>* Classifying is identifying the types or groups of something, showing how each category is distinct from the others.</li></ul> <p><b>Creative Thinking – 2 marks</b></p> <p>Creative thinking is expansive, open-ended invention and discovery of possibilities. When people speak of "right brain" activity, they most often mean creative thinking. Here are some of the more common creative thinking abilities:</p> <p>b) <ul style="list-style-type: none"><li>* Brainstorming ideas involves asking a question and rapidly listing all answers, even those that are far-fetched, impractical, or impossible.</li><li>* Creating something requires forming it by combining materials, perhaps according to a plan or perhaps based on the impulse of the moment.</li><li>* Questioning actively reaches into what is unknown to make it known, seeking information or a new way to do something.</li></ul></p> <p><b>Communicating – 2 marks</b></p> <ul style="list-style-type: none"><li>* Analyzing the situation means thinking about the subject, purpose, sender, receiver, medium, and context of a message.</li><li>* Choosing a medium involves deciding the most appropriate way to deliver a message, ranging from a face-to-face chat to a 400-page report.</li><li>* Evaluating messages means deciding whether they are correct, complete, reliable, authoritative, and up-to-date.</li><li>* Following conventions means communicating using the expected norms for the medium chosen.</li></ul> | <p><b>10</b></p> |
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|---|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
|   |    | <p>* Listening actively requires carefully paying attention, taking notes, asking questions, and otherwise engaging in the ideas being communicated.</p> <p>* Reading is decoding written words and images in order to understand what their originator is trying to communicate.</p> <p><b>Collaborating – 2 marks</b></p> <p>* Allocating resources and responsibilities ensures that all members of a team can work optimally.</p> <p>* Brainstorming ideas in a group involves rapidly suggesting and writing down ideas without pausing to critique them.</p> <p>* Decision-making requires sorting through the many options provided to the group and arriving at a single option to move forward.</p> <p><b>Example – 2 marks</b></p> |    |
| 2 | a) | <p><b>Mentioning the Process – 2 marks</b></p>  <p><b>Example – 2 marks</b></p> <p><b>Briefly explaining the Process – 6 marks</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 10 |
|   | b) | <p>Briefly explaining the below mentioned - <b>5 marks</b></p> <ol style="list-style-type: none"> <li>Immersion</li> <li>Research</li> <li>Observations</li> <li>Interviews</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 5  |



|   | a) | Listing 10 logical questions to ask teachers regarding problems faced while using chalk and black board – <b>1x10 = 10 marks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>10</b> |                                                                                                                               |         |                                                                       |                                                                                                                        |                                                                      |                                                                                                            |                                                 |                                                                                                               |                                                                                  |                                                                                                                               |
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| 3 | b) | NEEDS (FINDINGS): User’s emotional or physical necessities and desires that come in form of verbs.<br>INSIGHTS: Remarkable realizations that could be leveraged to better respond to a design challenge. – <b>2 marks</b><br>Differentiating with example – <b>3 marks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>5</b>  |                                                                                                                               |         |                                                                       |                                                                                                                        |                                                                      |                                                                                                            |                                                 |                                                                                                               |                                                                                  |                                                                                                                               |
|   |    | <table><tr><th>FINDING</th><th>INSIGHT</th></tr><tr><td>Community members drive to other locations for food and entertainment</td><td>Community members drive to other locations for food and entertainment because there are not many options in their area</td></tr><tr><td>Community members like to visit places of relaxation during weekends</td><td>Community members like to visit places of relaxation during weekends as they are exhausted by work/ travel</td></tr><tr><td>Very few Community members have been vaccinated</td><td>Very few Community members have been vaccinated because of fear, as there was fake news spreading in the area</td></tr><tr><td>Community members are traveling to stay in other locations during covid lockdown</td><td>Community members are traveling to stay in other locations during covid lockdown as there are very few hospitals in this area</td></tr></table> |           | FINDING                                                                                                                       | INSIGHT | Community members drive to other locations for food and entertainment | Community members drive to other locations for food and entertainment because there are not many options in their area | Community members like to visit places of relaxation during weekends | Community members like to visit places of relaxation during weekends as they are exhausted by work/ travel | Very few Community members have been vaccinated | Very few Community members have been vaccinated because of fear, as there was fake news spreading in the area | Community members are traveling to stay in other locations during covid lockdown | Community members are traveling to stay in other locations during covid lockdown as there are very few hospitals in this area |
|   |    | FINDING                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |           | INSIGHT                                                                                                                       |         |                                                                       |                                                                                                                        |                                                                      |                                                                                                            |                                                 |                                                                                                               |                                                                                  |                                                                                                                               |
|   |    | Community members drive to other locations for food and entertainment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           | Community members drive to other locations for food and entertainment because there are not many options in their area        |         |                                                                       |                                                                                                                        |                                                                      |                                                                                                            |                                                 |                                                                                                               |                                                                                  |                                                                                                                               |
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